

CBCS SCHEME



BCO601

Sixth Semester B.E./B.Tech. Degree Examination, June/July 2025 Microcontrollers and Embedded Systems

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the RISC design philosophy and how it differs from CISC architecture and List out its advantage with data flow.	10	L2	CO1
	b.	Discuss the concept of pipelining in ARM processors. How do interrupts effects the pipeline execution explain with one example.	10	L2	CO1
OR					
Q.2	a.	With the help of neat supporting Block diagram explain ARM Cox dataflow model.	10	L2	CO1
	b.	Explain the role of registers and also explain 4 fields of Current Program Status Register (CPSR).	10	L2	CO1
Module – 2					
Q.3	a.	List out all types of data processing institutions in ARM instruction set. Provide examples to illustrate the explanation.	10	L3	CO2
	b.	i) Write a program to add an array of 16-bit numbers and store the 32 bit result in internal RAM. ii) Write a program to find the largest / smallest number in an array of 32 numbers.	10	L3	CO2
OR					
Q.4	a.	Write a C program to find square of the number between (-1 – 10) and convert the same using assembly level program.	10	L3	CO2
	b.	i) Write assembly level program to interface a stepper motor and rotate it in clockwise and anticlockwise direction. ii) Write a assemble level program to find factorial of number.	10	L3	CO2
Module – 3					
Q.5	a.	Explain the different purpose of Embedded system with examples.	10	L2	CO3
	b.	Write a assembly level code and structural representation to display hex digits 0 to F on a 7 segment LED interface with an appropriate delay M in between.	10	L3	CO3
OR					
Q.6	a.	Explain different communication interface for embedded system with neat diagram.	10	L2	CO3
	b.	With the help of interfacing diagram write a assembly level program to interface a 4 x 4 keyboard and display the key code on an LCD.	10	L3	CO3
Module – 4					
Q.7	a.	Explain the quality attributes of embedded system with different types.	10	L2	CO4
	b.	Explain state machine model with two example : i) FSM model for Automatic Tea / Coffee vending machine. ii) FSM model for coin operated telephone system.	10	L2	CO4
1 of 2					

OR

Q.8	a.	Explain : i) Sequential Program model ii) Concurrent / Communicating Process model	10	L2	CO4
	b.	Explain automotive communication buses and key players of automotive embedded market concepts.	10	L2	CO4
Module – 5					
Q.9	a.	Explain the concept of task process threads with the help of neat diagram.	10	L2	CO5
	b.	Explain the concept of Dead lock and Dining Philosopher's problem.	10	L2	CO5
OR					
Q.10	a.	Explain message passing concept with neat diagram.	10	L2	CO5
	b.	What is semaphore explain binary semaphore concept with supporting diagram.	10	L2	CO5

Sixth Semester B.E. Degree Examination Microcontroller & Embedded Systems

TIME: 03 Hours

Max. Marks: 100

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.

Module -1			*Bloom's Taxonomy Level	Marks
Q.01	a	What are the RISC and CISC machine? Explain the major design rules that are implemented with RISC machine.	L2	8
	b	Briefly explain the various fields in current program status register.	L2	6
	c	List the differences between microcontroller and microprocessor.	L1	6
OR				
Q.02	a	Explain ARM based embedded system hardware components	L2	10
	b	What is pipeline in ARM? Explain the pipeline stages of ARM7, ARM9 & ARM10.	L2	10
Module-2				
Q. 03	a	Develop an ALP program to find the sum of 20 integer numbers.	L3	10
	b	Explain about Load and store Instructions in ARM with an example	L2	10
OR				
Q.04	a	Explain the barrel shifter operation in ARM processor with diagram. Illustrate with example for logical left shift operation.	L2	10
	b	Explain the following instructions with syntax and example: (i) MOV (ii) BIC (iii) RSB	L2	6
	c	Write a note on Profiling & Cycle Counting.	L1	4
Module-3				
Q. 05	a	Explain the classification of Embedded System.	L2	10
	b	With a neat diagram, Explain the interfacing of stepper motor through the driver circuit to microcontroller.	L2	10
OR				
Q. 06	a	What is an Embedded system? Explain any four purpose of Embedded system.	L2	10
	b	Write a note on : (i) Sensors and actuators (ii) I2C Bus in an Embedded system.	L1	10
Module-4				
Q. 07	a	Explain the Characteristics of Embedded System in detail.	L2	10
	b	Explain the role of Application specific integrated circuit (ASIC) on embedded system design.	L2	10
OR				
Q. 08	a	What are the operational & nonoperational quality attributes of Embedded systems?	L1	10
	b	Explain the Fundamental issues in Hardware Software Co-design	L2	10
Module-5				
Q. 09	a	With a neat diagram, explain operating System Architecture.	L2	10
	b	Explain the concept of Deadlock with an Example.	L3	10
OR				

Q. 10	a	Define Task, Process & threads. Explain process structure, process states and state transitions.	L2	8
	b	Briefly Explain the Role Integrated development Environment (IDE) for embedded software development.	L2	7
	c	Write a note on Message Passing	L1	5

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.

MODULE- 1

1. Explain the major design rules to implement RISC philosophy, State differences between RISC And CISC processors.
2. Discuss the ARM design philosophy, also discuss about the ARM bus technology
3. Explain ARM core data flow model with a neat diagram
4. Along with neat diagram of an ARM based embedded device (Microcontroller), explain the four main hardware components, Briefly Explain ARM register used under various modes
5. Explain Pipeline in detail.
6. Briefly describe the concept of exceptions, interrupts and the vector table
7. Describe conditional execution. Write the different code suffix.
8. Compare and Contrast microprocessor and microcontroller. 4M
9. Explain ARM core data flow model with a neat diagram. 8M
10. Along with neat diagram of an ARM based embedded device (Microcontroller), explain four main hardware components. 8M
11. Explain the different processor modes provided by ARM7. 8M
Give schematic of a Current Program Status Register of ARM7 processor briefing the individual bits.
12. What s Pipelining. Explain in detail schematically. 6M

1. What is barrel shifter? Describe the various operations barrel shifter supports?
2. Explain load store instructions with examples.
3. Explain different data processing instructions in ARM.
4. Explain the various Multiply instructions of ARM processor.
5. What are the various logical instructions supported by ARM? Explain each with examples
6. Explain the different branching instructions of ARM processor
7. Summarize the scheduling of Load instructions.
8. Summarize the scheduling of common instruction classes on ARM7TDM1.
9. Explain the Program Status Register (PSR) in ARM and list its fields. How can PSR be accessed and modified?
10. Explain how you would use Profiling and Cycle Counting to optimize an ARM assembly program. Give an example.

MODULE- 3

Passing Module

1. Explain the working principle with a diagram for each:
 - i) Memory sensors
 - ii) Actuators
 - iii) 7 segment LED display
 - iv) Stepper motor
 - v) Keyboard
 - vi) Push button switch
2. Explain embedded framework and classification of embedded system
3. What is embedded system? List its purposes & applications.
4. Differentiate between General purpose computing system & embedded system.
5. Explain program memory classification.
6. List different purposes of embedded system with examples.
7. Explain the following: i. I2C ii. 1-Wire Interface iii. SPI Interface iv. Reset Circuit

MODULE- 4

- 1. Explain the characteristics of embedded systems that differentiate them from general-purpose systems.**
- 2. Discuss the operational quality attributes of embedded systems with suitable examples.**
- 3. Explain the non-operational quality attributes of embedded systems. How do they influence the product's life-cycle and commercial success?**
- 4. Compare application-specific and domain specific embedded systems. Explain with one example for each category.**
- 5. Explain the concept of hardware-software co-design. Why is co-design important in embedded system development? Illustrate the typical co-design process.**
- 6. What is program modeling in embedded systems? Discuss any one common model (DFG, FSM, or Statecharts) with a diagram and example.**
- 7. Explain the process of designing an embedded system starting from specification to implementation. Highlight the role of co-design and modeling at each stage.**

MODULE- 5

Passing Module

1. Explain basics of operating system?

Explain types of operating system in embedded systems?

2. Differentiate between soft and hard real time?

3. Explain structure of process?

4. What is RTOS? Explain deferent services of RTOS?

5. Explain in detail task, process and thread?

Also Explain multitasking concept with an example.

6. What is scheduling? Explain the various for scheduling algorithms in RTOS?

7. Explain task communication & task synchronization?

8. What is semaphore? Explain various types of semaphores in RTOS?

9. Explain how to choose an RTOS in Embedded system?

10. Explain integration of hardware and firmware design of embedded system.

11. Explain detail Embedded system development environment?